Project Proposal

CMPT310 - D200

Group 20

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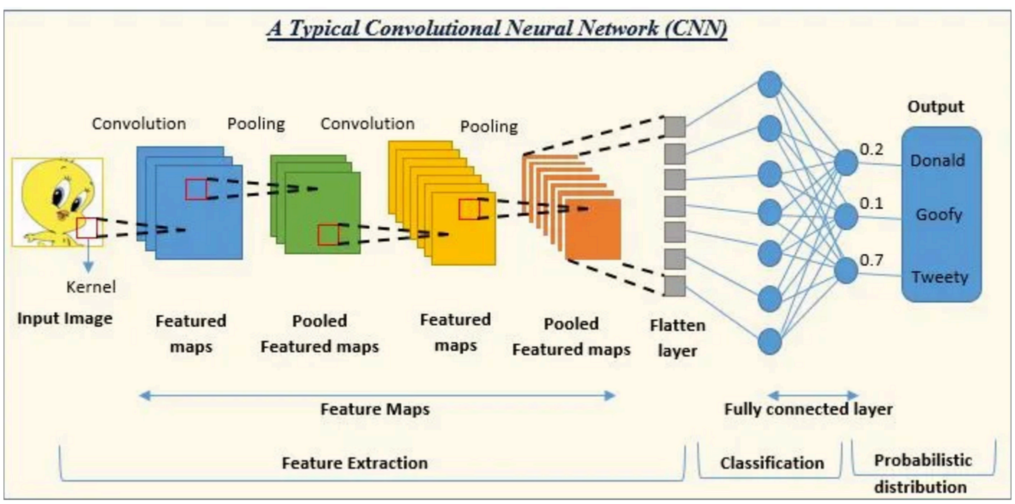
Project Idea:

We will develop an AI model capable of performing image categorization to identify 50 dog breeds with – images. We decided to do the 50 most popular breeds. The AI is going to identify common features in each breed like fur color, texture, etc with visual computing. The problem we aim to solve is users not being able to determine the breed of their dogs, this can be applied to general purposes because knowing a dog's breed is useful for determining the best type of food, what are its traits and instinct, and even for veterinary context. All of this in an automated way.

The project will take images of dogs as a raw image file JPG or PNG as input. This will go through the processing of feature extraction using different image filters / kernels and then output the 3 most likely matches for the dog's breed giving their a percentage of similarity. We decided to output the 3 most likely breeds because there are breeds that are really similar in features increasing the error of determining an incorrect breed. The AI techniques used will be classified using a Convolutional Neural Network (CNN) that extracts the spatial features of the image file (such as edges, textures, and colors) so that way we can determine the 3 most likely breed probabilities. To prevent overfitting, we will use Batch Normalization.

Image input -> CNN -> 3 most probable matches outputted [WIP]



Tools and Resources:

We will use tools like PyTorch, OpenCV, NumPy & Scikit-learn are going to be the main tools that are going to be used for the development of the AI classifier.

- PyTorch: Building and training the CNN
- OpenCV: Image manipulation, color space and file inputs
- NumPy & Scikit-learn: Mathematics, models and matrices.
- We are willing to adapt to different tools if the project requires it.

Any data you'll use (e.g., UCI dataset, synthetic images, simulation data)

- We are using an already labeled dataset from Stanford with more than 120 breeds and 20580 images of dogs. See appendix A for the 50 chosen breeds..
- Labeled Database: http://vision.stanford.edu/aditya86/ImageNetDogs/
- We plan on doing a split of the 50 breeds on this data for training/testing, which is 75% images for training and 25% for testing
- We will use a pre-processed database, we may add our own data during testing or even training to increase our probabilities.

Project Plan / Timeline

Milestone 1: March 1

- At least get the image loading and filtering ready using either OpenCV or NumPy for the convolutions
- Hopefully we can finish the convolutional and pooling layers before starting on the actual neural network

Milestone 2: March 29

- Hopefully we can finish the neural network and test it with the test images from the data set and also with some of our own pictures

Minimal Viable System

The simplest working version of the system takes in an image of a dog, extracts its features and tries to classify it.

Our core priority is implementing the first few CNN layers to process images and detect features such as edges

In the most basic form, our system could be viable to tell whether an animal is a dog or not.

Appendix A

Top 50 most popular dogs:

https://www.akc.org/most-popular-breeds/

https://www.countryliving.com/life/kids-pets/g3283/the-50-most-popular-dog-breeds/

French Bulldog, Labrador Retriever, Golden Retriever, German Shepherd, Siberian Husky, Rottweiler, Boxer, Doberman, Great Dane, Pug, Whippet, Newfoundland, Samoyed, Vizsla, Papillon, Beagle, Chihuahua, Pomeranian, Shih-Tzu, Yorkshire Terrier, Border Collie, Bernese Mountain Dog, Shetland

Sheepdog, Maltese, Collie, Rhodesian Ridgeback, West Highland White Terrier, Basset Hound, Afghan hound, Borzoi, Saluki, Irish wolfhound, Scottish deerhound, Great Pyrenees, Saint Bernard, Tibetan mastiff, Kuvasz, Malamute, Ibizan hound, Mexican hairless, Komondor, Eskimo dog, Dingo, Leonberg, Old English sheepdog, Otterhound, Bloodhound, Giant schnauzer, Gordon setter, Weimaraner